

Excel tables and conditional formatting

Excellent! Since you're learning **Excel for Data Analyst**, these are two of the most important topics. I'll explain them **step by step**, with examples and when to use them.

1. Excel Tables

What is a Table?

An Excel Table is a range of data that Excel treats as one organized object.

Instead of working with normal cells, Excel gives you powerful features like:

- Sorting
- Filtering
- Automatic formulas
- Automatic formatting
- Dynamic ranges

Imagine this data:

Employee ID Name Department Salary

101	Ravi	HR	30000
102	Priya	IT	50000
103	Arun	Finance	45000

This is normal data.

If you convert it into an Excel Table, Excel adds many extra features automatically.

How to Create a Table

Method 1

1. Select all your data.
2. Press **Ctrl + T**
3. Tick **My table has headers**
4. Click **OK**

Done!

Method 2

Select data

↓

Insert Tab

↓

Table

↓

OK

Why do we use Tables?

Because tables make data easier to manage.

Instead of writing formulas again and again,

Excel automatically copies formulas.

It also expands when you add new rows.

Features of Tables

1. Automatic Headers

Example

| ID | Name | Salary |

Headers stay fixed while scrolling.

2. Automatic Filters

Every column gets a dropdown.

You can:

- Sort A to Z
- Sort Z to A
- Filter specific values
- Search data

Example

Only show IT employees.

Just click the filter.

3. Automatic Formatting

Excel gives colors automatically.

Example

Blue rows

White rows

Alternate row colors

This improves readability.

4. Automatic Formula Fill

Example

Salary Bonus

30000 =Salary*10%

When you press Enter,

Excel automatically fills the formula for every row.

No need to drag.

5. Expands Automatically

Suppose your table has 10 employees.

Now you type employee number 11 below.

Excel automatically includes it in the table.

No need to recreate anything.

6. Structured References

Instead of

=B2*10%

Excel writes

=[@Salary]*10%

This means

Current row Salary

It is easier to understand.

Example

Normal Formula

=D2*10%

Table Formula

=[@Salary]*10%

Both give the same answer.

Changing Table Design

Click anywhere inside the table.

Go to

Table Design

You can

- Change colors
 - Add total row
 - Remove filters
 - Rename table
-

Total Row

Click

Table Design

↓

Tick

Total Row

Excel automatically calculates

- Sum
- Average
- Count
- Maximum
- Minimum

Example

Salary

30000

40000

50000

Total Row

Sum = 120000

Removing a Table

Table Design

↓

Convert to Range

Your data stays.

Only the table features disappear.

Advantages of Tables

- ✓ Easy filtering
 - ✓ Easy sorting
 - ✓ Automatic formulas
 - ✓ Better formatting
 - ✓ Dynamic data
 - ✓ Used in Power BI
 - ✓ Used in Dashboards
 - ✓ Used in Data Analysis
-

Real Data Analyst Example

Suppose every day HR sends employee data.

Instead of changing formulas every day,

You just paste the new data.

The table updates automatically.

Very useful.

2. Conditional Formatting

What is Conditional Formatting?

Conditional Formatting changes the appearance of cells **automatically based on rules or conditions**.

Instead of checking data manually,
Excel highlights important values.

Example

Marks

Student Marks

A	95
B	40
C	78
D	22

Suppose

Marks above 80
should become Green.

Marks below 35
should become Red.

Excel does this automatically.

Why do we use Conditional Formatting?

To quickly identify

- High values
- Low values
- Duplicate data
- Missing values
- Deadlines
- Errors

- Trends

Where is it?

Home Tab

↓

Conditional Formatting

Types of Conditional Formatting

1. Highlight Cell Rules

Most commonly used.

Examples

Greater Than

Less Than

Between

Equal To

Text Contains

Duplicate Values

Example 1

Salary

30000

45000

80000

Rule

Greater than

50000

Result

80000 becomes Green.

Example 2

Marks

20

55

90

Rule

Less than

35

Result

20 becomes Red.

Example 3

Duplicate Values

Names

Ravi

Priya

Ravi

Arun

Rule

Duplicate Values

Result

Both Ravi cells become highlighted.

Useful for finding duplicate records.

2. Top/Bottom Rules

Example

Top 10 Salaries

Excel highlights

Highest salaries.

Bottom 10

Highlights

Lowest salaries.

Above Average

Highlights

Values above average.

Below Average

Highlights

Values below average.

Example

Marks

40

50

70

90

Average = 62.5

Above average

70

90

Highlighted.

3. Data Bars

Instead of colors,

Excel shows bars inside cells.

Example

Sales

100

300

700

1000

The biggest value gets the longest bar.

Useful for comparing values visually.

4. Color Scales

Excel uses colors to show value ranges.

Example

Low Sales → Red

Medium Sales → Yellow

High Sales → Green

Very useful for dashboards.

5. Icon Sets

Excel adds icons.

Example

Sales Target

95% 

70% 

30% 

Common icons include:

- Arrows
 - Traffic lights
 - Stars
 - Flags
-

New Rule

You can create your own condition.

Example

Highlight salary

Greater than

Average Salary + 10000

Or use formulas for advanced rules.

Formula-Based Conditional Formatting

Suppose you want to highlight employees whose salary is greater than ₹50,000.

Data:

Name Salary

Ravi 30000

Priya 60000

Arun 45000

Select the salary cells (B2:B4), then create a new rule using this formula:

=B2>50000

Result:

Only **60000** is highlighted.

Managing Rules

Go to:

Home

↓

Conditional Formatting

↓

Manage Rules

Here you can:

- Edit rules
 - Delete rules
 - Change priority
 - Apply rules to different ranges
-

Clear Rules

To remove formatting:

Home

↓

Conditional Formatting



Clear Rules

Real Data Analyst Examples

Sales Dashboard

- Sales > ₹1,00,000 → Green
- Sales < ₹20,000 → Red

Attendance Report

- Attendance < 75% → Red

Employee Data

- Duplicate Employee IDs → Yellow

Inventory

- Stock < 10 → Red

Project Tracking

- Due date within 3 days → Orange

Difference Between Tables and Conditional Formatting

Excel Tables

Organizes data

Supports sorting and filtering

Automatically expands with new rows

Makes formulas easier

Conditional Formatting

Highlights important data

Uses colors, icons, and bars

Changes appearance based on conditions

Makes patterns and issues easy to spot

Interview Questions

1. What is an Excel Table?

An Excel Table is a structured range of data with built-in features like sorting, filtering, automatic formulas, and dynamic expansion.

2. What are the benefits of Excel Tables?

Automatic formatting, filters, structured references, auto-filled formulas, and easy data management.

3. What is Conditional Formatting?

A feature that automatically changes the appearance of cells based on specified conditions.

4. Give some uses of Conditional Formatting.

Highlight high or low values, find duplicates, identify missing data, track deadlines, and visualize trends with color scales or data bars.

Practice Exercise

Create this data in Excel:

ID	Name	Department	Salary
----	------	------------	--------

101	Ravi	HR	30000
-----	------	----	-------

102	Priya	IT	60000
-----	-------	----	-------

103	Arun	Finance	45000
-----	------	---------	-------

104	Neha	IT	75000
-----	------	----	-------

105	Ravi	HR	30000
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Then:

1. Convert the data into an **Excel Table** (Ctrl + T).
2. Filter to show only the **IT** department.
3. Add a **Total Row** to calculate the total salary.
4. Use **Conditional Formatting** to:
 - Highlight salaries greater than **₹50,000** in green.
 - Highlight duplicate names.
 - Apply **Data Bars** to the Salary column.
 - Apply a **Color Scale** to the Salary column.

Mastering these two features will make working with large datasets much faster and is a skill commonly expected in Data Analyst interviews.